THE ORDER-ZERO MEAN-VALUE FORMULA IN H^p

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AI assistance was used in adapting and checking the argument and the exposition.

Brevig and Kouroupis state: "If f in $\mathscr{H}^2$ has an analytic continuation to $\mathbb{C}_0$ of order 0, then

$$\|f\|_{\mathscr{H}^2}^2 = \lim_{\sigma\to 0^+} \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |f(\sigma+it)|^2\,\mathrm{d}t.$$

" Corollary 2, §2. The inferred question asks whether this ordered formula extends to every $1 \le p < \infty$.

Order zero means $\mu_f(\sigma) \le 0$ for every $\sigma > 0$, where $\mu_f(\sigma) = \limsup_{|t|\to\infty} \log|f(\sigma + it)|/\log|t|$; the proposed H^p formula replaces the squared norms and integrands by their pth powers.

The source is Brevig and Kouroupis, *Mean values, order, and convergence in the Hardy space of Dirichlet series*, Corollary 2, arXiv:2609.35496; the shift theorem is Carando, Defant, Marceca, Schoolmann, and Sevilla-Peris, *Hypercontractivity and strips of convergence in Hardy spaces of general Dirichlet series*, Corollary 1.2, arXiv:2401.09996.

Answered: yes. Evidence: Sketch.

Conditional on: arXiv:2609.35496, Theorem 1, and arXiv:2401.09996, Corollary 1.2.

Yes, conditional on the source's H^2 mean-value theorem and the hypercontractivity theorem for ordinary Dirichlet series.

Proof. Let $k = \lceil p/2 \rceil$. First suppose $g \in \mathscr{H}^{2k}$ has an order-zero continuation. Choose Dirichlet polynomials $P_j \to g$ in $\mathscr{H}^{2k}$. The function $(g - P_j)^k$ belongs to $\mathscr{H}^2$ and has order zero. Brevig–Kouroupis' Corollary 2 therefore gives, for every $\sigma > 0$,

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |g(\sigma+it) - P_j(\sigma+it)|^{2k}\,\mathrm{d}t = \|(g-P_j)(\sigma+\cdot)\|_{\mathscr{H}^{2k}}^{2k}.$$

Minkowski's inequality and $L^p \le L^{2k}$ on the normalized interval compare the p-mean of g with that of P_j. The latter converges to $\|P_j(\sigma+\cdot)\|_{\mathscr{H}^p}$ by the mean theorem for polynomials; letting $j \to \infty$ proves

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |g(\sigma+it)|^{p}\,\mathrm{d}t = \|g(\sigma+\cdot)\|_{\mathscr{H}^p}^{p}.$$

For a general $f \in \mathscr{H}^p$, fix $\sigma > 0$ and choose $0 < \varepsilon < \sigma$. Ordinary-frequency hypercontractivity implies $f_\varepsilon(s) = f(s+\varepsilon) \in \mathscr{H}^{2k}$; its continuation still has order zero. Apply the preceding result to f_ε at $\sigma - \varepsilon$. This gives the inner mean identity for f at every $\sigma > 0$. Finally, horizontal translations are contractive and $f_\sigma \to f$ in $\mathscr{H}^p$ as $\sigma \downarrow 0$, first for polynomials and then by density. Taking the outer limit proves the ordered formula. □

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